Toward Detecting Conspiracy Language in Misinformation Documents

Alana Platt
University of Wisconsin-Whitewater, Whitewater, Wisconsin
USA
platta@uww.edu

Jonathan Brown
University of Wisconsin-Whitewater, Whitewater, Wisconsin
USA
brownjc98@uww.edu

Amanda Venske
University of Wisconsin-Whitewater, Whitewater, Wisconsin
USA
venskeal20@uww.edu

ABSTRACT

Recent events have illustrated the danger of online conspiracy theories to cause harm in the real world. In this research in progress, we focus on machine learning techniques to differentiate conspiracy information from other forms of misinformation. Our results demonstrate that conspiracy language is differentiable and suggest there may exist features of conspiracy language independent of the conspiracy topic. We also present a direction for future work to better understand the unique features of conspiracy language and how they may be used to enhance machine learning techniques.

CCS CONCEPTS

• Human-centered computing → Collaborative and social computing → Empirical studies in collaborative and social computing; Information systems → World Wide Web → Web mining

KEYWORDS

Misinformation detection, conspiracy detection, text mining, conspiracy communities

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1 INTRODUCTION

The proliferation of misinformation on social media, websites, and other online spaces is a topic of concern for both researchers and the general public alike. Misinformation can take many forms, and can be driven by a variety of motivations, ranging from the innocent resharing of false or misleading information to sophisticated misinformation campaigns promulgated by bad actors. Conspiracy theories and associated information have recently received much attention in the media and general public.

While conspiracy theories themselves are not necessarily misinformation, they are often accompanied by misleading or outright false claims, statistics, and other supporting material that are misinformation. They also have the potential for significant negative consequences outside the realm of the internet. For example, QAnon conspiracy theorists were among the mob who violently stormed the US Capital building on January 6, 2021[17], and anti-vaccine conspiracies have led to lower vaccination rates during the Covid-19 pandemic [21].

In this research project, we propose to extend the current work on general misinformation by demonstrating that machine learning can (1) differentiate conspiracy language from among a corpus of misinformation and (2) identify characteristics indicative of conspiracy language as opposed to other misinformation documents.

2 PRIOR WORKS

Misinformation can be thought of as false or misleading information, with disinformation using the false information in the effort to purposely deceive [10]. Departing from traditional news source veracity, fake news utilizes dishonest intentions to mislead readers [7, 20]. Conspiracy theories further develop down the track of disinformation with engaging in a proposed element of secrecy to explain events and/or actions of conspirators allegedly acting in their own best interest [22]. Like many human-driven information sources, conspiracy theories cover many domains of thoughts. In recent years, geopolitical, environmental, and public health conspiracy theories [3], particularly those relating to the COVID-19 pandemic [6], have become especially prevalent. Additionally,

the spread of information in verified communications regarding time-sensitive public health issues can vary, as seen when exploring the efficacy of Twitter as a mode of crisis communications for public health departments during the COVID-19 pandemic [11].

2.1 Foundations

Without standard terminology, incorrect information, including conspiracy theories, has ill-defined characteristics. Misinformation is a major concern across areas of scholarship [1, 10]. Our particular area of focus is detecting conspiracy theories. Detecting conspiracy theories holds a sense of urgency, due to the harm they can cause [14] and the speed at which they can spread [4]. Human nature intertwines with conspiracy theories to result in chaotic shaping of communication. Some have theorized that people choose to endorse conspiracy theories due to the paranoia that arises out of "mass opinion" [15]. This further develops due to how information is available. With "fake news" proliferating on social media without credibility, mainstream news organizations are nearly seen as gatekeeping the information spread [2, 13]. Conspiracy believers have also been found to believe in the conspiracy even after being presented with credible debunking from mainstream and independent outlets [12]. Conspiracy theories are now strongly ingrained in modern society [15, 22], further establishing the importance of understanding their features. The social factors of conspiracy communities cannot be ignored [16]. Conspiracy theories spur collective action, and current conspiratorial thinkers encourage those on the fringes of conspiratorial beliefs to delve deeper into those social groups.

2.2 Detection

Detection centered on misinformation is building more robust lines of work. Fake news detection often needs to utilize linguistic- and visual-based content, as well as social context [20]. Recognizing the power and potential harm of social media in the spread of fake news, Shu et al. establish the distinctions between how detection needs to be approached based on the difference between fake news in traditional media sources and fake news on social media. Extracting features of source, headline, body text, and image/video for news content sources and user-, post-, network-based features for social context, the authors propose a model with two main distinctions: knowledge-based and style-based. Key takeaways include comparing and contrasting different types of misinformation, which shows the applicability of knowledge-based and style-based models to rumor classification, truth discovery, and clickbait and spam detection [20]. The authors assert applicability by highlighting discrepancies and inconsistencies in newly posted social media and news articles, suggesting the ability to characterize fake news diffusion and provide proactive fake news intervention. Other proposed models for fake news detection [8] start with automated feature extraction and move to multi-source, weighted vector combination. Recognizing that fake news is often not a binary of fake and not fake, the authors classified fake news into multiple degrees of fakeness. They successfully showed a model distinguishing different classes of fake news, with hopes to more finely classify different degrees of fakeness in the future. Other work focuses more specifically on systems for rumor detection, like the Hoaxy platform [19]. The Hoaxy platform's goal was to build a uniform platform to collect and track misinformation and fact checking. Pulling from both social networks and news sites and mainly utilizing tweets, they tracked tweet volume and examined correlation with social media activity and fake news site activity. Key findings suggest that the platform's design could be useful for studying the relationship between social media use and fake news. Possible solutions after detecting need to effectively counter misinformation, with specific user-centered response [9] Kirchner and Reuter conducted surveys and semi-structured interviews and found most social media users preferred approaches which clearly included disclaimers about the false information. This suggests that effective future strategies to fight misinformation should follow-through with marking articles accordingly. Our particular focus on conspiracy theories adds to a line of work that seeks to specify certain features of conspiracy theories that can contribute to better understanding the nature of their discussion in online communities. Since user activity varies in different periods of information spread, rumor components can be identified in useful patterns [5]. Of particular note is how identified conspiratorial statements signal narrative-motif through recurring Agent-Action-Target triplets [18]. Samory and Mitra set out to identify topics of discussion in conspiracy spaces on Reddit, as well as recurring narrative elements. Using an approach they call "Bag-of-Infinitygrams", they determined motifs that suggest what conspiratorial thinkers discuss and how they talk about conspiracy theories. Computing a document-term matrix following Bag-of-Words processes, the authors used infinitygrams instead of words. Infinitygrams were retained to use in their "Bag-of-Infinitygrams" process if they appeared at least 20 times in the corpus, were delimited by word boundary characters, and contain at least one non-stopword.

3 DATASET & METHODS

The data used from this study were a set of websites identified by the OpenSources [1] projects having problematic content. Specifically, it contains a set of websites compiled by media professionals that are categorized as at least one of the following: conspiracy, bias, political, clickbait, hate, satire, junksci, fake, unreliable, rumor, and blog. According to OpenSources, all websites were given category labels by analyzing six areas of information for each website: the domain title, "About Us" section, cited sources, writing styles, aesthetics and screen design, and social media presence.

Our dataset was collected by downloading all accessible sites in that list in June 2021. Some of the websites were still active, while others were no longer being updated. Some were no longer accessible. We analyzed the downloaded sites and removed those that were not primarily text based (e.g. podcast/video oriented

[1]h https://libguides.njstatelib.org/facts/fake_news

content, link aggregator sites). We were able to scrape and retain 161 of the 833 websites on the list, resulting in 3807 articles. Out of 161 websites, 63 were labeled conspiracy. We preprocessed the data by removing HTML tags, removed stopwords, and special characters, performed stemming and generated n-grams.

4 RESULTS

In order to test our first research question to determine if machine learning can differentiate between conspiracy language from among a corpus of misinformation, we trained and tested several classifiers as discussed in Section 4.1. Further, discuss our next steps in identifying unique characteristics of conspiracy language in Section 4.2.

4.1 Detecting Conspiracy Documents

We trained several models using several machine learning algorithms and empirically determined that Multinomial Naive Bayes performed best, with an F1-Score of 80.62%. We then began to refine our model. We first hypothesized that applying sentiment analysis to distinguish between strongly positive and negative emotions could provide further differentiation between conspiracy information and other types of misinformation; however, empirical results did not support this. We then reviewed our highest quality words ranked by their *term frequency – inverse document frequency* (TF-IDF). We observed that many of the documents contained similar topics (e.g. gun control) regardless of the presence of conspiracy information. Thus, we hypothesized that the differentiator in detecting conspiracy information may come from how the topic is discussed rather than the overall topic itself. To test this, we proposed removing the highest rated TF-IDF terms and retaining most of the mid-range TF-IDF terms. The intuition is that the highest TF-IDF terms would correspond to the overall topic of the document, while the mid-range TF-IDF terms would be important to tell us how that topic was being discussed. We modified our model to remove these terms in 5% increments, and the results are shown in Table 1.

Table 1: Misinformation detection performance

% removed	Precision	Recall	F1
0%	74.78%	92.29%	80.62%
5%	79.65%	91.09%	84.99%
10%	88.04%	89.10%	88.57%
15%	89.14%	88.43%	88.78%
20%	90.50%	87.37%	88.91%
25%	91.44%	86.70%	89.01%
30%	92.36%	85.24%	88.66%

As shown in Table 1, removing the top TF-IDF terms improves the performance of the classifier, with Precision increasing dramatically (16.66%) with a comparatively modest decrease in recall (5.59%) peaking with the removal of the top 25% of terms and a corresponding F1 measure of 89.01%.

4.1 Toward Identifying Conspiracy Features

In the next phase of this research in progress, we will explore the characteristics that differentiate conspiratorial language from other types of misinformation. Our results in section 4.1 provide support for this hypothesis, and it requires further inquiry. One avenue of research is to further investigate the specific types of terms and how they differ between conspiracy and non-conspiracy documents on the same topic.

Recognizing the growth of conspiratorial thinking in the subreddit r/conspiracy [18], we plan to work with data from Reddit community for additional insights. With over 1.6 million subscribers, this would allow a current snapshot of conspiratorial thinking and indicate potential terms of conspiratorial thinking. The subreddit experienced a rapid increase in users in recent years, as Samory and Mitra cite that r/conspiracy had over 500K users at the time of their 2018 research on the language of conspiracies in online communities [18]. This data will augment our existing dataset by allowing us to investigate conspiratorial language among a community. Specifically, we want to explore unique characteristics of conspiracy language, including: common terms and motifs in conspiracy language; unique patterns of information seeking; community specialized community language (e.g. puns such as "Fraudci" or unique spellings like "Cl🤡wn W🌎rld"); and other features interesting features. Our overall aim is to use these insights to further enhance machine learning detection of conspiracy language in documents.

5 DISCUSSION & CONCLUSIONS

In this research in progress, we demonstrate that machine learning techniques can differentiate between conspiracy language and other forms of misinformation. Further, our experiments provide evidence of a language of conspiracy that exists beyond the primary topic of the conspiracy, as evidenced by the improved performance of models with up to 25% of the top TF-IDF terms removed. A number of open questions exist as avenues of inquiry into the nature of conspiracy language as opposed to other misinformation documents, including the next steps for this research in progress are outlined in Section 4.2. Additionally, we will further explore an observed relationship between the conspiracy category and categories of junk sci, bias, and fake.

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